



SAFETY DATA SHEET

Synergex

SECTION 1. PRODUCT IDENTIFICATION

Product name : Synergex

Other means of identification : Not applicable

Recommended use : Antimicrobial Food Surface Treatment

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0,11 % - 0,2 %

Company : Manufactured or Distributed in Costa Rica by:
ECOLAB Costa Rica SRL, Zona Franca BES, El Coyoil 5 Km al Oeste
Aeropuerto Juan Santamaría,
Alajuela, Costa Rica, Código Postal: Apdo109-4003.

Product Distributed in Honduras by:
ECOLAB Ofibodega Premier, bodega No.8 en el Boulervar del este a
30 mts de la caseta de peaje, San Pedro Sula, Honduras

Product Distributed in Panama by:
ECOLAB ALBROOK COMERCIAL PARK, L. D14, CALLE RUBEN
DARIO, CURUNDU, PANAMÁ

Product Distributed in El Salvador by:
ECOLAB, S.A, de C.V. 12 Calle Poniente y 23 Avenida Sur #700 San
Salvador, El Salvador Colonia Cucumayan.

Product Distributed in Guatemala by:
ECOLAB S.A. ECOLAB 42 CALLE 23-00 ZONA 12, BODEGA 8

Product Distributed in Nicaragua by:
ECOLAB C.C.R.L . ECOLAB KM. 16.7 CARRETERA A
TICUANTEPE, 200 MTS A COMUNIDAD SAN PEDRO,
TICUANTEPE. - MANAGUA, NICARAGUA

Emergency telephone : Costa Rica: For medical emergency information, call 911 and (506)
2436-3900 in Costa Rica. National Center of Intoxication in Costa
Rica: (506) 2223-1028
El Salvador: National Center of Intoxication in EL SALVADOR TEL.
(503) 22319262
Guatemala: TOXICOLOGICAL INFORMATION CENTER OF
GUATEMALA CIAT (502) 2230 0807
Nicaragua: National Center of Intoxication in NICARAGUA N° Tel
(505) 2897150
Honduras: HOSPITAL ESCUELA DE TEGUCIGALPA HONDURAS
(504) 2232-2322
Panama: SYSTEM OF EMERGENCY MANAGEMENT OF THE
REPUBLIC OF PANAMA (SUME) - TEL. 911

Issuing date : 13.10.2021

SECTION 2. HAZARD OR HAZARDS IDENTIFICATION

GHS Classification
Product AS SOLD

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Oxidizing liquids, Category 3	H272
Acute toxicity (Oral), Category 5	H303
Skin corrosion/irritation, Category 1	H314
Serious eye damage/eye irritation, Category 1	H318
Specific target organ toxicity - single exposure, Category 3, Respiratory system	H335
Short-term (acute) aquatic hazard, Category 3	H402
Long-term (chronic) aquatic hazard, Category 3	H412

Product AT USE DILUTION

Skin corrosion/irritation, Category 1	H314
Serious eye damage/eye irritation, Category 1	H318

GHS label elements

Product AS SOLD

Hazard pictograms



Signal Word : Danger

Hazard Statements : H272 May intensify fire; oxidizer.
 H303 May be harmful if swallowed.
 H314 Causes severe skin burns and eye damage.
 H335 May cause respiratory irritation.
 H412 Harmful to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
 P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
 P220 Keep away from clothing and other combustible materials.
 P261 Avoid breathing mist or vapors.
 P264 Wash skin thoroughly after handling.
 P271 Use only outdoors or in a well-ventilated area.
 P273 Avoid release to the environment.
 P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
 P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.
 P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.
 P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
 P312 Call a POISON CENTER/ doctor if you feel unwell.
 P363 Wash contaminated clothing before reuse.
 P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

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P405

Store locked up.

Disposal:

P501

Dispose of contents/ container to an approved waste disposal plant.

Product AT USE DILUTION

Hazard pictograms

:



Signal Word

: Danger

Hazard Statements

: H314

Causes severe skin burns and eye damage.

Precautionary Statements

: **Prevention:**

P264

Wash skin thoroughly after handling.

P280

Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.

P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.

P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor. Wash contaminated clothing before reuse.

P363

Storage:

P405

Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Product AS SOLD

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

SECTION 3. COMPOSITION / INFORMATION OF HAZARDOUS INGREDIENTS

Product AS SOLD

Pure substance/mixture

: Mixture

Chemical name	CAS-No.	Concentration (%)
Sulfuric acid	7664-93-9	10 - 30
Hydrogen peroxide	7722-84-1	10 - 30
Secondary Alkanesulphonates	5324-84-5	5 - 10
Acetic acid	64-19-7	5 - 10
Peroxyacetic acid	79-21-0	1 - 5
Octanoic acid	124-07-2	1 - 5
9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts	1315321-94-8	1 - 5
Peroxyoctanoic acid	33734-57-5	0,1 - 1

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Product AT USE DILUTION

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

Product AS SOLD

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.
- Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.
- Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Potential Health Effects

Product AS SOLD

- Eyes : Causes serious eye damage.
- Skin : Causes severe skin burns.
- Ingestion : May be harmful if swallowed. Causes digestive tract burns.
- Inhalation : May cause respiratory tract irritation. May cause nose, throat, and lung irritation.

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Chronic Exposure : Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns.

Ingestion : Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

Chronic Exposure : Health injuries are not known or expected under normal use.

SECTION 5. FIRE-FIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : None known.

Specific hazards during fire fighting : Special protective equipment for fire-fighters
Oxidizer. Contact with other material may cause fire.
Oxidizer; material is an oxidizer which may readily react with other materials, especially upon heating.

Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
Sulfur oxides

Special protective equipment for fire-fighters : In case of fire, wear a full face positive-pressure self contained breathing apparatus and protective suit.

Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

SECTION 6. MEASURES TO BE TAKEN IN CASE OF ACCIDENTAL SPILLAGE

Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Isolate the waste do not allow it to come into contact with incompatible materials. For small spills contain with sand or vermiculite and dilute the contained product at least 10 times with water. Transfer to an open topped container and remove to a safe place for neutralization* / disposal. For large spills contain spill and evacuate the area, leave until the reaction subsides, then collect up for disposal. Obtain consent from the local water company / authority if

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considering discharge to sewer. *NEUTRALIZATION : once diluted, neutralize with a suitable alkali such as sodium bicarbonate. Combustible materials exposed to this product should be rinsed immediately with large amounts of water to ensure that all product is removed. Residual product which is allowed to dry on organic materials such as rags, cloths, paper, fabrics, cotton, leather, wood, or other combustibles may spontaneously ignite and result in a fire.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures	: Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling	: Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
Conditions for safe storage	: Keep in a cool, well-ventilated place. Keep away from reducing agents. Keep away from strong bases. Keep away from combustible material. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers. Pressure bursts may occur due to gas evolution if the container is not adequately vented.
Storage temperature	: 5 °C to 35 °C

Product AT USE DILUTION

Advice on safe handling	: Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
Conditions for safe storage	: Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

SECTION 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

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Components	CAS-No.	Form of exposure	Permissible concentration	Basis
Sulfuric acid	7664-93-9	TWA	0,2 mg/m3	CR OEL
Sulfuric acid	7664-93-9	TLV-TWA	0,2 mg/m3	DO OEL
Sulfuric acid	7664-93-9	CPT	1 mg/m3	PA OEL
		CCT	3 mg/m3	PA OEL
Sulfuric acid	7664-93-9	TWA (Thoracic particulate matter)	0,2 mg/m3	ACGIH
		TWA	1 mg/m3	NIOSH REL
		TWA	1 mg/m3	OSHA Z1
Hydrogen peroxide	7722-84-1	TWA	1 ppm	CR OEL
Hydrogen peroxide	7722-84-1	TLV-TWA	1 ppm	DO OEL
Hydrogen peroxide	7722-84-1	CCT	2 ppm 3 mg/m3	PA OEL
		CPT	1 ppm 1,5 mg/m3	PA OEL
Hydrogen peroxide	7722-84-1	TWA	1 ppm	ACGIH
		TWA	1 ppm 1,4 mg/m3	NIOSH REL
		TWA	1 ppm 1,4 mg/m3	OSHA Z1
Acetic acid	64-19-7	TWA	10 ppm	CR OEL
		STEL	15 ppm	CR OEL
Acetic acid	64-19-7	CPT	10 ppm 25 mg/m3	PA OEL
		CCT	15 ppm 37 mg/m3	PA OEL
Acetic acid	64-19-7	TWA	10 ppm	ACGIH
		STEL	15 ppm	ACGIH
		STEL	15 ppm 37 mg/m3	NIOSH REL
		TWA	10 ppm 25 mg/m3	NIOSH REL
		TWA	10 ppm 25 mg/m3	OSHA Z1
Peroxyacetic acid	79-21-0	STEL (Inhalable fraction and vapour)	0,4 ppm	CR OEL
Peroxyacetic acid	79-21-0	STEL	0,4 ppm	ACGIH

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

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- Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

- Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

- Eye protection : Safety goggles
Face-shield
- Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
- Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing
- Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Physical State	: liquid	liquid
Color	: clear, colorless	colorless
Odor	: pungent	odorless
pH	: 0,2 - 1,5, (100 %)	1,5 - 3,0
Flash point	: Not applicable	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: > 35 °C	
Evaporation rate	: No data available	
Flammability (solid, gas)	: Not applicable	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: 1,195 - 1,215	
Water solubility	: soluble	

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Solubility in other solvents	: No data available
Partition coefficient: n-octanol/water	: No data available
Autoignition temperature	: No data available
Thermal decomposition	: No data available
Viscosity, kinematic	: No data available
Explosive properties	: No data available
Oxidizing properties	: yes
Molecular weight	: No data available
VOC	: No data available

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Contamination may result in dangerous pressure increases - closed containers may rupture.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: Direct sources of heat. Exposure to sunlight.
Incompatible materials	: Mild steel Metals Polyvinyl chloride
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides Sulfur oxides

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion

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Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Toxicity

Product AS SOLD Product

Acute oral toxicity	: Acute toxicity estimate : 3.769 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : > 40 mg/l Test atmosphere: vapor
Acute dermal toxicity	: Acute toxicity estimate : > 5.000 mg/kg
Skin corrosion/irritation	: No data available
Serious eye damage/eye irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Product AS SOLD

Environmental Effects	: Harmful to aquatic life with long lasting effects.
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Product AT USE DILUTION

Environmental Effects	: This product has no known ecotoxicological effects.
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Product AS SOLD Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: 48 h EC50 Daphnia magna (Water flea): 82 mg/l
Toxicity to algae	: No data available

Components

Toxicity to fish	: Sulfuric acid 96 h LC50: 22 mg/l Acetic acid 96 h LC50 Oncorhynchus mykiss (rainbow trout): > 1.000 mg/l Peroxyacetic acid 96 h LC50: 0,8 mg/l
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Octanoic acid
96 h LC50 *Lepomis macrochirus* (Bluegill sunfish): 22 mg/l

9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts
96 h LC50: 0,75 mg/l

Peroxyoctanoic acid
96 h LC50 Fish: 0,15 mg/l

Components

Toxicity to algae : Hydrogen peroxide
72 h EC50: 1,38 mg/l

Acetic acid
72 h EC50 *Skeletonema costatum* (marine diatom): > 1.000 mg/l

Peroxyacetic acid
72 h EC50: 0,7 mg/l

9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts
72 h EC50: 2,56 mg/l

Persistence and degradability

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. INFORMATION REGARDING THE DISPOSAL OF PRODUCTS

Product AS SOLD

Disposal methods : Avoid that the product entering in the body of water, drains, pipes, or the earth (grounds). Where possible, recycling is preferable to disposal or incineration in accordance with local legislation. If you can't recycle, dispose according to local regulations. Elimination of waste in approved waste disposal plants.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations. Remove as unused products. Empty containers should be brought to local recycling or waste disposal according to local legislation. Do not reuse empty containers.

Product AT USE DILUTION

Disposal methods : Avoid that the product entering in the body of water, drains, pipes, or the earth (grounds). Where possible, recycling is preferable to disposal or incineration in accordance with local legislation. If you can't recycle, dispose according to local regulations. Elimination of waste in approved waste disposal plants.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

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Remove as unused products. Empty containers should be brought to local recycling or waste disposal according to local legislation. Do not reuse empty containers.

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

UN number : 3149
Description of the goods : HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE, STABILIZED
Class : 5.1 (8)
Packing group : II
Environmentally hazardous : no

Air transport (IATA)

Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

UN number : 3149
Proper shipping name : HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE, STABILIZED
Class : 5.1 (8)
Packing group : II
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code : Not applicable
Marine pollutant : no

SECTION 15. REGULATORY INFORMATION

Our Safety Data Sheet (SDS) complies with Decree n ° 40705-S

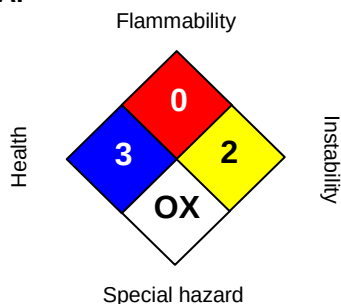
SECTION 16. OTHER INFORMATION

Product AS SOLD

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NFPA:



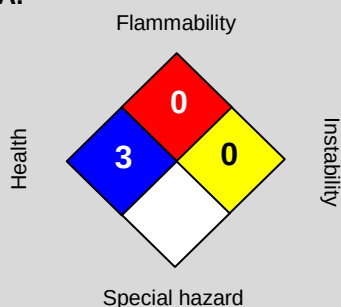
HMIS III:

HEALTH	3
FLAMMABILITY	0
PHYSICAL HAZARD	2

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

NFPA:



HMIS III:

HEALTH	3
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 13.10.2021
Version : 1.1
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.